

Relationship Between Additional Employment Stability Fee and Foreign Workers in Taiwan

1. Introduction

〈1〉Research Background

In 1989, Taiwan established a system for importing foreign laborers, marking the first introduction of foreign workers to engage in low-skilled jobs. With Taiwan's population aging, the demand for caregiving services has steadily increased. Starting in 1992, Taiwan began allowing foreign caregivers and domestic helpers to work in the country. By the year 2022, statistics from the Directorate-General of Budget, Accounting and Statistics of the Executive Yuan showed that the majority of Taiwan's foreign workers hail from Southeast Asian countries such as the Philippines, Indonesia, and Vietnam. They are primarily employed in manufacturing and caregiving industries.

The introduction of foreign labor has undoubtedly had a significant impact on Taiwan's labor market, providing substantial labor supply in a relatively short period. Wage regulations and various policies have influenced the labor market from multiple perspectives, indirectly altering the structure of Taiwan's labor market.

Distinct cultural and language differences have given rise to complex issues, including integration with employers, assessment of foreign workers' rights, and the applicability of Taiwan's Labor Standards Act. These issues have spurred ongoing discussions within Taiwan, reflecting the high degree of attention and concern surrounding the topic domestically.

〈2〉Research Objectives

While numerous studies have traditionally emphasized the impact of introducing foreign labor on the domestic workforce, this study aims to shift attention towards the effects of Taiwan's government policies on foreign laborers themselves.

This research article specifically investigates the implications of the Employment Stability Fee and Additional Employment Stability Fee that employers are required to pay for foreign

workers. The study seeks to understand how these policies influence the wages of foreign laborers and the mechanisms through which such impacts manifest in the labor market.

Key objectives include:

- 1. **Impact on Foreign Laborer Wages:** Analyzing how the implementation of ESF and AESF policies affects the wage levels of foreign workers in Taiwan.
- 2. **Labor Market Responses:** Exploring the reactions and adjustments within the labor market in response to these policy measures, including potential shifts in employment patterns, labor demand, and employer behaviors towards foreign workers.

By focusing on these aspects, the study aims to provide insights into the direct effects of Taiwan's regulatory framework on foreign laborers, contributing to a nuanced understanding of policy impacts within the broader context of labor market dynamics.

2.Policy and Economic Theory

〈1〉Policies Regarding Foreign Migrant Workers

Before delving into the topic of Employment Stability Fee, it's important to briefly introduce the "3K Five-Level System" in the manufacturing industry, as it bears significant relevance to policies concerning foreign migrant workers. Implemented by the government in 2010, this system applies to 105 specified industries characterized by abnormal temperatures, dust, toxic gasses, organic solvents, chemical treatments, non-automated operations, and other designated processes.

These industries are categorized into five levels (A+, A, B, C, D) based on their specific production characteristics. One of the major distinctions among these levels is the allocation ratio for foreign migrant workers, which varies as follows: A+ (35%), A (25%), B (20%), C (15%), and D (10%).

	A+	A	B	C	D
highest rate	35%	25%	20%	15%	10%

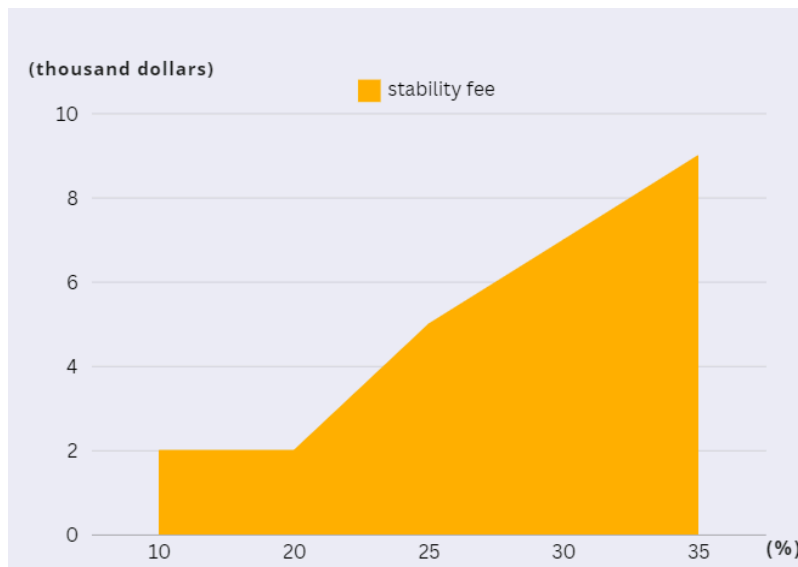
This categorization directly influences the hiring and deployment of foreign workers within these industries, reflecting the government's strategy to manage and regulate labor utilization based on industry-specific risks and requirements. Understanding this system is crucial for contextualizing how policies like the Employment Stability Fee interact with regulations governing foreign labor in Taiwan's manufacturing sector.

Starting from 2013, Taiwan implemented an additional mechanism known as the Employment Stability Fee, which depends on the "3K Five-Level System" allocation rates. Employers are required to pay an extra fee if they exceed the maximum allowable allocation rate for their industry grade when hiring foreign workers. This policy directly influences decisions regarding foreign labor employment and contributes significantly to labor cost management within these sectors.

<i>extra rate</i>	<i>original fee</i>	<i>extra fee</i>	<i>total</i>
5%	\$2000	\$3000	\$5000
5%-10%	\$2000	\$5000	\$7000
10%-15%	\$2000	\$7000	\$9000

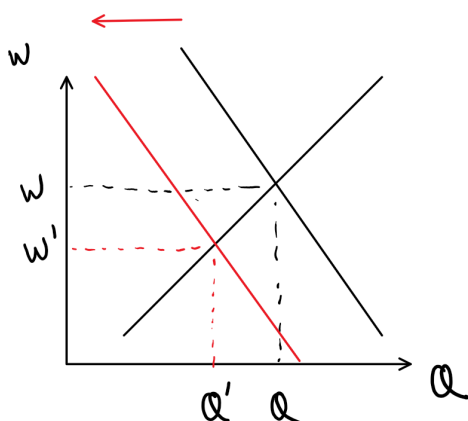
The additional employment allocation rate can go up to an extra 20%, with the exception of the trading industry, which can go up to 40%. For example, if it belongs to the B-grade industry and wishes to employ 30% foreign workers, exceeding the maximum allocation rate by 10%, then for each additional foreign worker hired, a monthly additional Employment Stability Fee of 5,000 yuan is required (this 5,000 yuan includes the original stability fund of 2,000 yuan).

Below is a chart illustrating the total Employment Stability Fee, including the original 2,000 yuan and additional ESF, for industries with a maximum allocation rate of 20% when employing different percentages of foreign workers (unit: thousand yuan).

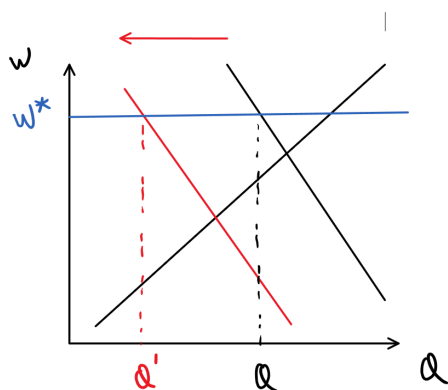


〈2〉Labor Market Supply and Demand

For employers, the Employment Stability Fee represents an additional cost they must pay to the government when hiring foreign workers, thereby increasing the overall cost of employing them. Consequently, it is anticipated that employers may reduce their hiring of foreign migrant workers. This would shift the demand curve for foreign labor to the left, resulting in a leftward movement of the equilibrium point of supply and demand. Both the wage and the number of foreign workers employed would decrease as illustrated below:



但若考慮到最低薪資的影響，原本外籍勞工市場不在完全競爭市場的均衡點上，若就業安定費造成需求線左移，不一定會造成薪資的降低，而只有雇用人數的下降，如下圖：



參、研究方法與步驟

一、資料來源

本文的實證資料，係來自勞動部「外籍勞工管理及運用調查」。此調查以經勞動部勞動力發展署核准僱用移工之事業面及家庭面雇主為調查對象。調查範圍為臺灣地區(含金門縣及連江縣)之製造業、營建工程業及醫療保健及社會工作服務業、其他服務業。調查表式分為「事業面移工調查表」及「外籍家庭看護工調查表」。抽樣方法採分層比例抽樣，事業面以行業、規模為分層變數；家庭面則以縣市別、國籍別為分層變數。調查方式採郵寄問卷，輔以電話催收。4451-5839。本研究取用100年至108年事業面資料，103年之《外籍勞工工作及生活關懷調查表》因缺乏研究需要之變數，故排除。共八年資料每年約4451至5839份樣本，共計40737份。

二、研究對象

因時間與心力限制，本研究僅取事業面資料中特定產業。選擇方式係先計算不同產業「申請外加就業安定費」之工廠家數占比，取比率最高與最低之產業，再取兩個資料較多之產業。(因100至104年資料中，沒有調查是否申請安定費之變數，因此這裡的計算僅取104至108年資料)

產業代碼	樣本數(104-108年)	申請家數	申請家數占比
4	314	80	0.255
9	35	9	0.257
11	344	107	0.311
5	191	60	0.314
12	100	32	0.320
13	473	153	0.323

21	4010	1358	0.339
20	534	181	0.339
19	203	73	0.360
8	480	176	0.367
2	100	38	0.380
16	828	318	0.384
23	552	214	0.388
17	7751	3013	0.389
25	562	224	0.399
15	839	348	0.415
22	730	303	0.415
18	698	293	0.420
14	2361	1006	0.426
10	251	107	0.426
3	1275	547	0.429
1	1605	699	0.436
24	415	181	0.436
7	519	252	0.486
26	99	NA	

最後選擇結果如下：

- (一)成衣及服飾品製造業(產業代碼4)，樣本數499。
- (二)機械設備製造業(產業代碼21)，樣本數6417。
- (三)金屬製品製造業(產業代碼17)，樣本數11584。
- (四)紙漿、紙及紙製品製造業(產業代碼7)，樣本數833。

三、變量之定義與測量

(一)依變量：薪資

以資料中調查各年6月份平均每名外籍勞工之「總薪資」計，其中包含經常性薪資與加班費。

(二)自變量

1.安定費:資料中並沒有直接與廠商支付之外加安定費項目,因此本研究透過各家廠商外籍移工之雇傭比例,與政府定之外加安定費規定以及各產業之法定核發率,計算計算各家廠商在每名外籍移工需支付多少平均安定費作為變量。以下為計算公式:

$$\text{平均每名外籍移工外加安定費} = \frac{\text{廠商為所有外籍移工支付之外加安定費總額}}{\text{外籍移工雇傭量}} \\ = \frac{\text{超過法定核發率5\%內之移工數} \times 3000 + \text{超過法定核發率5-10\%之移工數} \times 5000 + \dots}{\text{外籍移工雇傭量}}$$

資料處理時依各樣本分別計算過於複雜,因此用以下方式計算近似值:
(事先算出perc = 各家廠商外籍移工人數占比)

$$\text{平均每名員工外加安定費} = \begin{cases} 0 & \text{if } perc \leq 0.2 \\ 3000 \times (perc - 0.2) & \text{if } 0.2 < perc \leq 0.25 \\ 150 + 2000 \times (perc - 0.25) & \text{if } 0.25 < perc \leq 0.3 \\ 250 + 2000 \times (perc - 0.3) & \text{if } 0.3 < perc \leq 0.35 \\ 350 + 2000 \times (perc - 0.35) & \text{if } 0.35 < perc \leq 0.4 \end{cases}$$

$$\text{平均每名外籍移工外加安定費} = \frac{\text{平均每名員工外加安定費}}{perc}$$

(三)控制變量

- 1.工廠規模:並非以資料中之規模變數,因為僅區分了兩類,可能無法充分掌握對依變數之影響。故本研究以總員工人數作為規模變數。
- 2.產業類型:僅以挑選特定產業做不同回歸模型方式,並非新增產業虛擬變數。
- 3.年份:新增各年份虛擬變數。

四、分析方法

建立四個不同產業之多變量回歸模型,檢視外加安定費對外籍移工薪資之影響,並加以敘述統計分析。

肆、研究結果分析

以下為迴歸分析結果:

	成衣及服飾品製造業			紙漿、紙及紙製品製造業		
外加安定費	0.3101	(0.438)	***	0.20	(0.29)	
工廠規模	2.69	(0.7593)		11.98	(1.18)	***
year_101	1604	(537.5)	**	234.97	(450.10)	
year_102	545.4	(594.5)		-408.03	(491.02)	
year_104	1594	(533.1)	**	99.38	(432.99)	

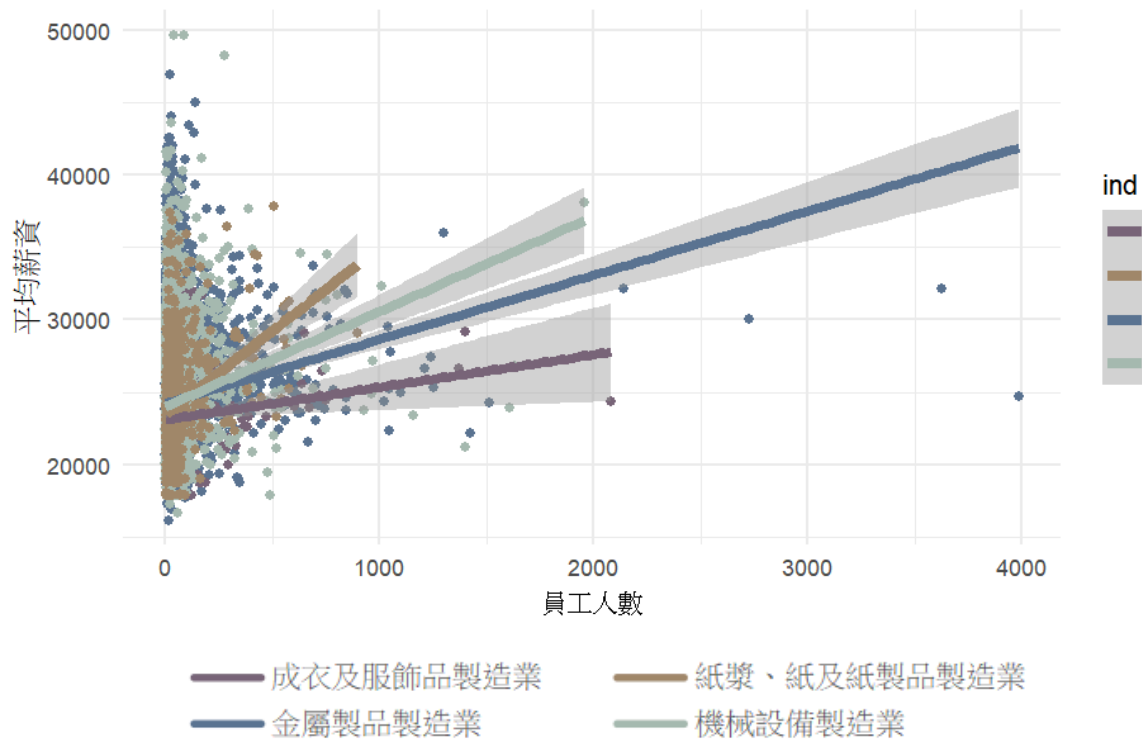
year_105	1573	(524.5)	***	651.69	(443.12)	
year_106	2043	(571.4)	***	1191.85	(452.96)	**
year_107	2865	(571.7)	***	1720.81	(450.26)	***
year_108	3685	(560)	***	2963.20	(456.00)	***
constant	21530	(398.9)		22903.71	(341.64)	***
Adjusted R-squared	0.1277			0.1728		
p-value	4.499E-12			< 2.2e-16		
N	499			833		

	金屬製品製造業			機械設備製造業		
外加安定費	0.4961	0.09034	***	-0.28	(0.14)	*
工廠規模	4.223	(0.3055)	***	6.14	(0.54)	***
year_101	1065	(136.6)	***	936.83	(174.68)	***
year_102	298.1	(146.3)	*	172.72	(193.37)	
year_104	1559	(129.6)	***	1425.74	(178.72)	***
year_105	1367	(128.7)	***	1461.14	(174.09)	***
year_106	2038	(133.1)	***	2129.70	(179.24)	***
year_107	2439	(135.7)	***	2949.75	(185.52)	***
year_108	3388	(134.4)	***	3551.54	(184.31)	***
constant	22790	(101.9)	***	22728.68	(135.83)	***
Adjusted R-squared	0.09683			0.1127		
p-value	< 2.2e-16			< 2.2e-16		
N	11584			6417		

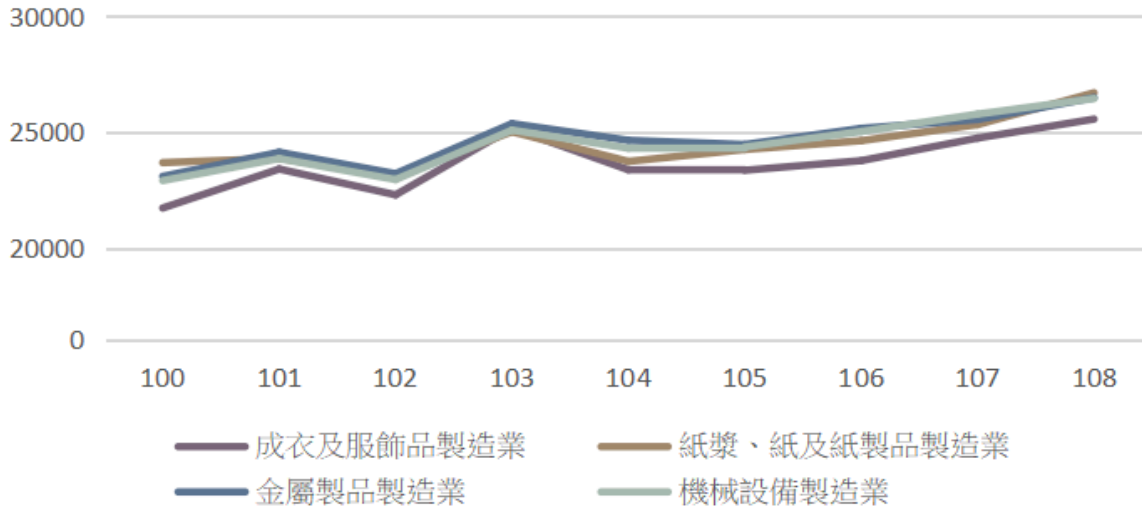
一、基本變量對薪資之影響

本研究挑選了四個產業，建立四個回歸模型。四個模型在各變數對薪資之影響不盡相同，將依序解釋。

從工廠規模變數觀察，成衣及服飾品製造業(以下簡稱成衣業)模型中，此項變數影響薪資之關係不顯著，另外三個產業則非常顯著。代表對此三個產業，越大規模的工廠傾向給外籍移工更高的薪資。



接著是時間變數，研究發現除了102年外，薪資大致上呈現逐年增長。在102年時，因為外加安定費進行調整，造成在所有產業的薪資出現整體性下跌。紙漿、紙及紙製品製造業在101至105年增長不顯著，可能因為其他外生衝擊，也可能因為樣本數較少。



二、安定費對薪資之影響

接著觀察安定費對薪資影響，在成衣業以及金屬製造業，安定費對薪資都是極度顯著的正相關，紙品製造業則不顯著，只有在機械製造業是顯著的負相關。這和我們先前所預測的有所出入，我們要進一步探討可能原因。

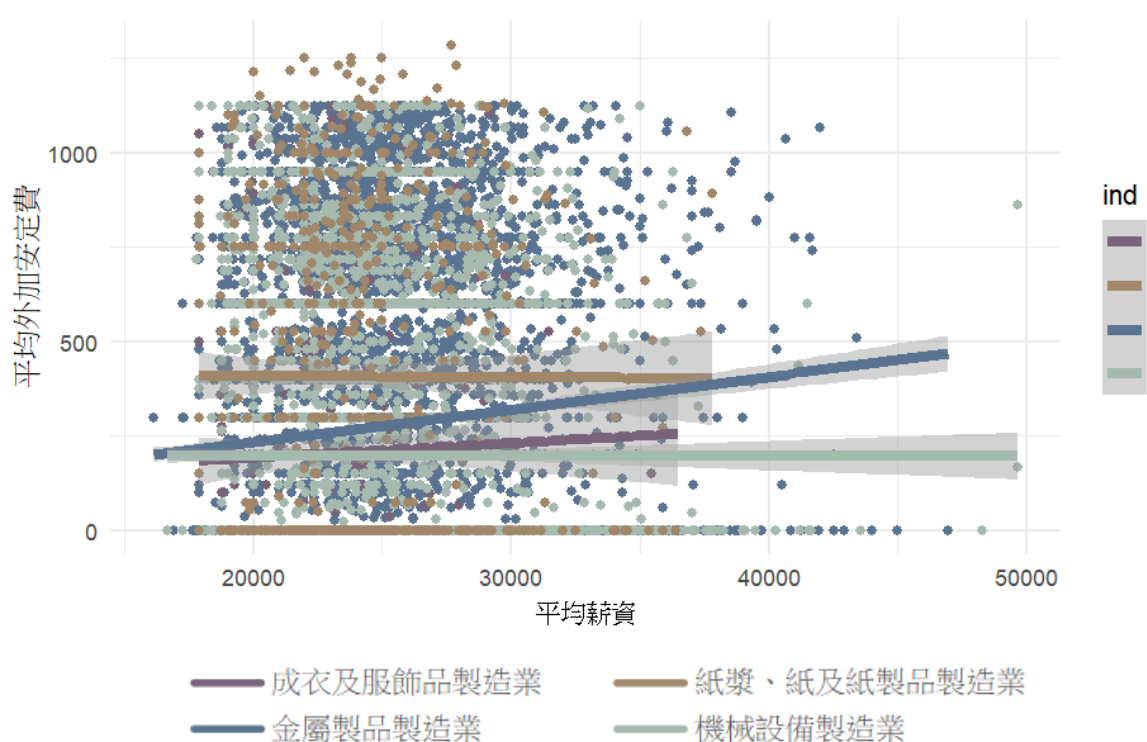
根據陳坤銘《引進外籍勞工對台灣經濟影響之再分析》研究，台灣之實質工資具相當程度之向下僵固性，外籍移工的影入對台灣勞工薪資的影響有限，同理在安定費較高時，雇用外

籍移工之廠商同樣無法降低外籍移工之薪資。這可能是使結果不為負相關之原因之一，但並非主因。

需支付越多安定費之廠商代表雇傭之外勞占比較高，也能合理推斷其對外勞之需求越強烈，較傾向以較高薪資聘用外勞，此內生性因素可能才是造成此結果的主因。並且我們也能推斷在成衣業與金屬製造業中，此內生效果較強，越缺工的廠商傾向支付較高工資，而在機械製造業此內生關係則較弱。

三、不同產業下之影響

在不同產業，就業安定費對於薪資的影響也不同，但就目前研究結果，無法完全掌握原因。從圖中能看到，唯一負斜率的是機械製造業(D)，在整體平均外加安定費也偏低，因為他屬於核發率低的產業。在金屬製品製造業(C)，缺工廠商傾向支付更多安定費關係明顯，可能是因為此產業廠商間附加價值差距價大，較賺錢的廠商仍能夠雇傭更多外勞。圖：



伍、結論與建議

1. 支付越高安定費的廠商，通常願意支付更高薪水吸引勞工

需支付較高安定費之廠商是因為他僱用較高的外勞比例，也代表此廠商更想以外籍勞工取代本勞，原因可能來自本勞薪資過高，或是供給不足。從課稅角度看，因為這些廠商需求彈性較低，所以能夠獲得較多稅收，符合稅收效率，但如果要以減少外籍移工雇傭量已釋出工作機會給本勞，效果就比較差。如果要以有效減少數量，就統一各個雇傭占比稅率，使得需求彈性高之廠商能因此改成雇傭本勞。

2. 安定費對於薪資的負面影響，通常小於廠商缺工之內生效果

從廠商的角度看，勞工之雇傭取決於邊際成本與邊際產出，雖然雇傭外勞之邊際成本因為安定費變高，但邊際產值較高的廠商會以較高薪資僱傭員工，低邊際產值廠商則不再雇傭外勞，就不需支付安定費。

3. 無法從本研究得知安定費率改變對整體市場之影響

因為本研究是在同一時期不同廠商需支付的安定費與薪資之關係，廠商之反應已經結束，我們從確定之現況分析無法排出不同廠商之特性，也就無法排除最重要之內升因素。這樣的分析也只能區分各家廠商或產業之間的不同，無法觀察到當費率改變時，整體市場之反應程度。

4. 無法得知安定費對外籍移工雇傭量之改變

本研究資料來源在樣本上沒有年度連續性，故無法得知安定費改變時，廠商如何調整外籍移工雇傭量。

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陸、參考資料

勞動部《外籍勞工管理及運用調查》

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行政院主計處資料